In the following by ultrafilter on X we mean an ultrafilter on $P(X)$. Denote τ_x the open neighbourouds of x. We say $U \in St(X)$ converges to $x \in X$ meaning that every open neighbouroud of x is in U.

Esercizio 1. *Let (X, τ) be a topological space. Then tfae:*

1. *X compact*
2. *$\forall U \in St(X)$ U converges to some $x \in X$*

Proof.

- 1) $\Rightarrow$ 2): By contradiction assume

$$\exists U \in St(X) \forall x \in X \exists V_x \in \tau_x V_x \notin U$$

  Let $C_x = X \setminus V_x$. Then $\forall x \in X$ $C_x \in U$. Observe that $\{V_x : x \in X\}$ is an open covering of X, so by compactness we can extract a finite sub-covering $\{V_{x_1}, .., V_{x_n}\}$. Then $\bigcap_{i=1,..,n} C_{x_i} = \emptyset$, contradiction since $\emptyset \notin U$.

- 2) $\Rightarrow$ 1): By contradiction assume $\{U_i : i \in I\}$ is some covering of X that does not admit a finite sub-covering. Denote $V_i := X \setminus U_i$ for all $i \in I$. This gives that $\{V_i : i \in I\}$ are closed subsets of X with the fip. Then there exists some ultrafilter U such that $\{V_i : i \in I\} \subseteq U$. Let $x \in X$ be convergence point for U. Since $\{U_i : i \in I\}$ is a covering of X there must be some j such that $x \in U_j$. But then $U_j \in U, V_j \in U$, which implies $\emptyset \in U$, contradiction.

□